ABSTRACT

AI-Assisted Search and Rescue Robot for Disaster Response

PROPOSED METHODOLOGY

Natural disasters such as earthquakes, building collapses, mine accidents, and fires put rescue teams at serious risk while survivors remain trapped. We propose a low-cost, AI-assisted Search and Rescue Robot that enters hazardous zones ahead of human responders to gather real-time data, enabling safer and faster rescue decisions.

The robot is built around an ESP32-CAM, which streams live video over Wi-Fi and serves as the central controller. An L298N motor driver powers four DC motors, letting the robot travel over rubble and uneven terrain. An MQ-2 sensor detects smoke and harmful gases, while a DHT11 sensor monitors ambient temperature. A buzzer and LED indicators provide immediate alerts in dangerous conditions.

SG90 SERVO MOTOR

What it does: Rotates the HC-SR04 ultrasonic sensor mounted on it, enabling obstacle detection.

Working Principle: A PWM signal from the ESP32 rotates the servo to a chosen angle, letting the robot "look" left and right before selecting a safe path.

PWM signal from ESP32 → Servo rotates to desired angle → e.g. 0°, 90°, 180°

The live camera feed is processed using AI-based human detection, helping rescue teams identify trapped victims quickly and accurately. Video and alerts are relayed to the rescue team's monitor to support fast, informed decisions. The entire system is designed to be affordable, portable, and easy to deploy in emergencies.

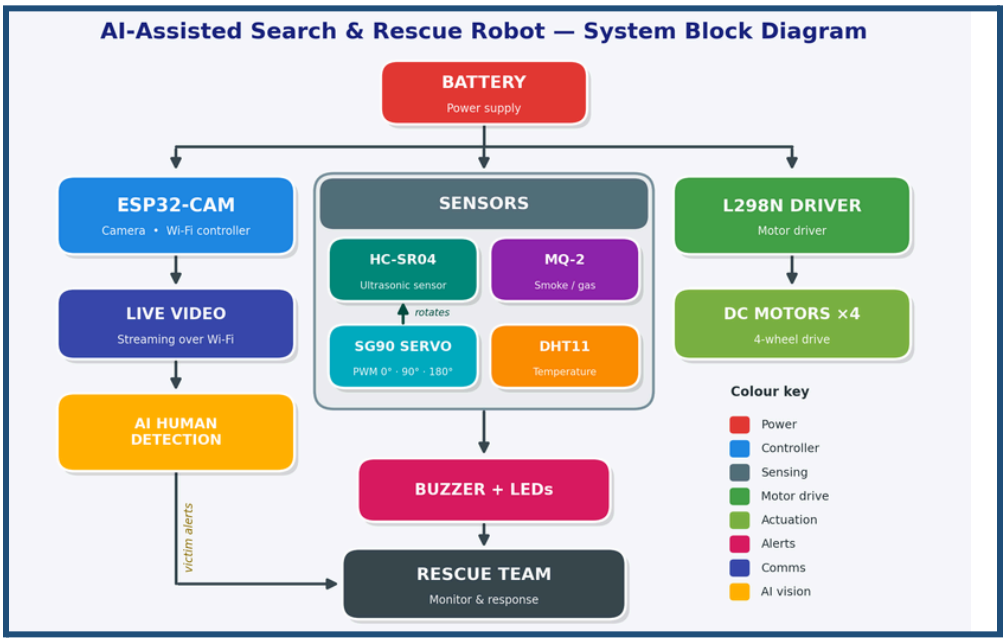


Fig. 1: Proposed block diagram of the AI-Assisted Search and Rescue Robot

PROPOSED OUTCOME

The robot delivers real-time visual monitoring, wide-angle obstacle detection, gas leakage detection, and temperature monitoring in hazardous environments, reducing risk to rescue personnel while increasing the speed and effectiveness of search operations. By integrating robotics, embedded systems, wireless communication, and AI-assisted victim detection, the project offers a practical, affordable, and socially impactful solution for modern disaster management and emergency rescue missions.